

Model Validation Report — Equity Selection Signals

Models under review	gbm (deployed Monte-Carlo ranking), gbm_expected (its closed-form expectation), momentum (12-1 benchmark rule), clam_orig / clam_2021 (CNN-LSTM-Attention forecaster)
Model owner	Long_Term_Trading / Quant_Model_Research (developer: the author, in an earlier role)
Validator	Signal_Validation — independent re-implementation; no model code modified or tuned
Data	3,000 US equities (top by market cap, snapshot 2026-09-15), daily prices 2013-01-02 → 2026-09-14, weekly rebalance 2015-01-02 → 2026-09-12 (612 weeks)
Report date	2026-09-15 (round 1) · 2026-09-16 (round 2 revalidation, §12)
Standard followed	SR 11-7 / OCC 2011-12 structure: purpose, data, methodology, results, findings, limitations, monitoring, conditions of use, decision

1. Executive summary and decision

model	Q1 overfit tests passed	Q2 current monitoring status	Q3 gate benefit (OOT)	decision
momentum	2 / 3 (fails Deflated Sharpe)	AUC 0.506, PSI 0.25 (watch)	max DD −29.7% → −22.0%, CAGR 35.5% → 23.7%	Conditionally approved as a benchmark only, under §9 conditions
gbm_expected	0 / 3	AUC 0.500, calibration slope −0.03	gate off 63% of weeks	Not approved — no evidence of skill
gbm (deployed)	0 / 3	AUC 0.502	gate off 92% of weeks	Not approved — implementation defect (F1)
clam_orig	—	—	—	Not approved — cannot be validated (F3, F4)
gbm_weekly (round 2)	2 / 3 (fails Deflated Sharpe)	AUC 0.507, calibration slope 0.74	gate destroys return (CAGR 19.3% → −2.6%)	Not approved — real but does not beat the momentum benchmark (§12)
clam_weekly_* (round 2)	0 / 3	AUC 0.50	gate off 61–92% of weeks	Not approved — two redevelopment variants, no evidence of skill (§12)

Across two rounds and eight candidate variants, none is approved for production use; the untuned momentum rule is retained as the benchmark that any future model must beat. The round-2 redevelopment (§12) confirmed that the GBM defect was an implementation error (fixed model passes the noise tests) and that the CLAM approach does not produce a usable weekly signal even after its data-construction defect is repaired. The most valuable outputs are five findings (§7) that came from reading and reproducing the deployed code rather than from any single statistic.

2. Purpose and scope

The model owner uses these signals to rank US equities weekly and hold the top names. This review asks the three questions a second-line function must answer before approving a model:

1. Is the backtested performance real, or an artefact of search and noise? (§5.1)
2. Is the signal still valid today, and how would we know when it stops being valid? (§5.2)

3. What is the rule for switching it off, and does that rule help out-of-time? (§5.3)

Out of scope: portfolio construction beyond equal-weight top-50, risk limits, and any change to the models themselves. The validator's job is to accept, reject, or condition — not to redevelop.

3. Models under review

model	as deployed	validator's reproduction
gbm	Fit μ, σ on 504 daily log returns; simulate one 65-step GBM path; score = $\text{mean}(\text{path})/S_0 - 1$	Ported verbatim from <code>stats_model_process.get_gbm_path_simulation</code> , fixed seed per date
gbm_expected	—	$E[\text{score}] = \text{mean}_i \exp(\mu \cdot t) - 1$: same model without the Monte-Carlo draw. Added by the validator to measure the draw's noise
momentum	—	$P[t-21] / P[t-252] - 1$. Textbook 12-1 momentum; no parameter was chosen by looking at the data
clam_orig	Keras CNN+LSTM+Attention, 252-day input of 5 log-diff features, 65-day output; score = $\exp(\sum \text{predicted dlog High}) - 1$	Weights and scaler loaded as shipped; inference code ported from <code>clam_inference.py</code>
clam_2021	—	Same training script, training window cut at 2021-12-31, to obtain an honest out-of-time period

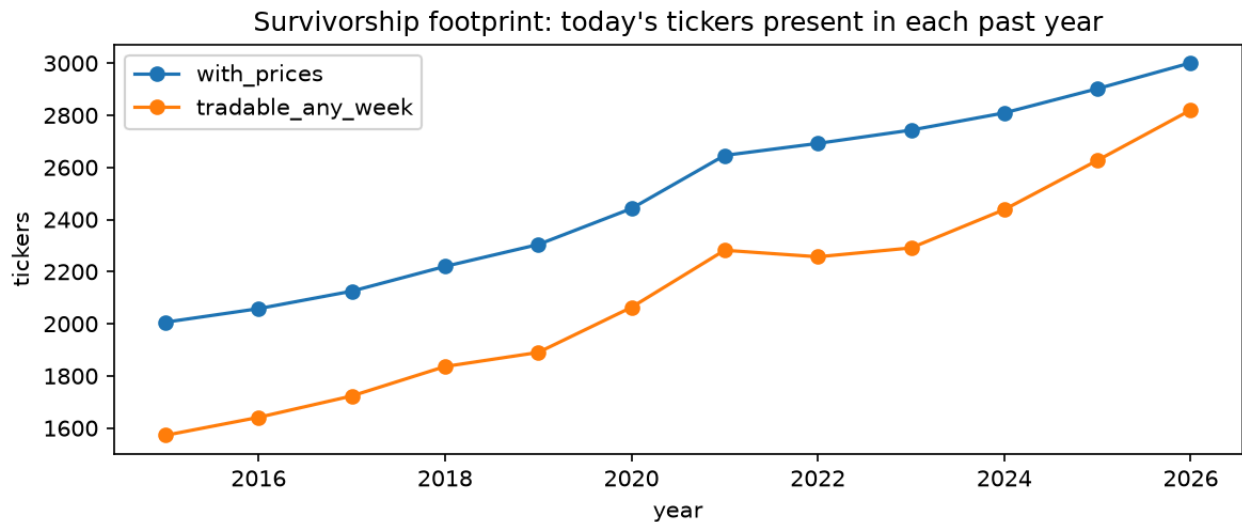
All models are evaluated with one backtest engine: weekly rebalance at Friday close, long-only, top-50 equal weight, 10 bps one-way cost on traded weight, liquidity filter applied before ranking. Performance is judged on **active return versus the equal-weighted tradable universe**, so that survivorship and universe effects shared by all candidates cancel.

4. Data

item	value	source
Universe	3,000 largest US common equities, one share class per company	Yahoo screener, snapshot 2026-09-15
Prices	8,104,194 daily OHLCV rows, 3,001 tickers (incl. SPY)	Yahoo Finance
Rebalance panel	1,492,706 (week, ticker) rows, 612 weeks	<code>sql/queries/build_panel.sql</code>
Tradable filter	close $\geq$ \$5 and 20-day average dollar volume $\geq$ \$5M; 74% of ticker-weeks pass	<code>config.py</code>
Storage	DuckDB; every derived table is reproducible from <code>sql/</code> and <code>sv/</code>	<code>sql/schema.sql</code>

Data quality. Three tickers (CBIO, NFE, DEC) carry zero or negative adjusted closes on 3,909 rows — a Yahoo dividend-adjustment artefact. They are kept in `prices` as received and excluded at the panel and feature layer. One rebalance week (2026-09-14) is a single-day partial week and carries no forward return.

Survivorship. Only 2,006 of today's 3,000 tickers have prices in 2015 (1,572 tradable); the count rises monotonically to 3,001 in 2026 (fig. 1). Companies delisted before the snapshot are absent. This inflates absolute returns for every candidate in the same direction; it is the reason all decisions are made on active return against the same universe, and it is listed as a limitation (§8).



5. Methodology

5.1 Q1 — overfitting tests (`sv/validation/overfit.py`)

test	guards against	statistic	pass rule
Stationary block bootstrap (Politis-Romano, mean block 13 weeks, 5,000 draws)	trusting a point estimate of Sharpe	95% CI of annualised active Sharpe	CI excludes 0
Deflated Sharpe Ratio (Bailey & López de Prado 2014), N = 3 trials	choosing the best of several tries; non-normal returns	P(true SR > expected max SR of 3 noise strategies)	≥ 0.95
Cross-sectional permutation null (500 shuffles of scores within each week)	the signal–return link being coincidental	share of permuted Sharpes $\geq$ observed, on gross returns	$p < 0.05$

The permutation null is evaluated on **gross** returns. A shuffled signal turns the book over $\sim 100\%$ a week; net of costs the null would sit at Sharpe ≈ -1.9 and any persistent signal would pass on cost savings alone. This was corrected during the review.

5.2 Q2 — monitoring (`sv/validation/monitoring.py`, `sql/queries/psi_score.sql`, `auc_by_year.sql`)

For each model and rebalance date, on the trailing 52 weeks ending the week before the date: `psi_score` (score distribution vs the first 104 weeks, 10 development-sample deciles), `psi_input` (trailing-252d realised vol, the regime variable), `auc_1w` / `ks_1w` (discrimination of next-week direction vs the cross-sectional median), `auc_13w` / `ks_13w` / `calib_slope` at the models' 65-day horizon (lagged 13 weeks so no target is used before it is known), `rolling_sharpe` and `active_drawdown` at portfolio level. PSI and AUC are recomputed per year in pure SQL as a cross-check.

5.3 Q3 — kill switch (`sv/validation/gate.py`)

A rule-based gate, thresholds fixed before the out-of-time window was examined:

rule	threshold	rationale
<code>psi_score</code>	> 0.25	industry convention for "population has shifted"
<code>rolling_sharpe</code>	< 0	trailing year of negative active return
<code>auc_1w</code>	< 0.50	no discriminatory power over the trailing year
<code>active_drawdown</code>	$< -15\%$	relative drawdown beyond tolerance

Any breach switches the model to cash for the coming week; each flip pays the full one-way cost. Portfolio-level metrics are lagged one week so the gate sees only information available on the decision date. Champion (ungated) and challenger (gated) are compared from 2022-01-03 (246 weeks).

6. Results

6.1 Full-sample backtest, 2015-01 → 2026-09

strategy	CAGR	vol	Sharpe	max DD	avg weekly turnover
momentum	27.0%	33.8%	0.88	−39.9%	15%
gbm_expected	18.8%	32.4%	0.70	−55.2%	10%
gbm (deployed)	3.9%	27.9%	0.28	−46.2%	90%
universe equal-weight	12.9%	20.4%	0.70	−40.2%	—
SPY	13.6%	16.6%	—	—	—

6.2 Q1 — overfitting tests (active return vs universe)

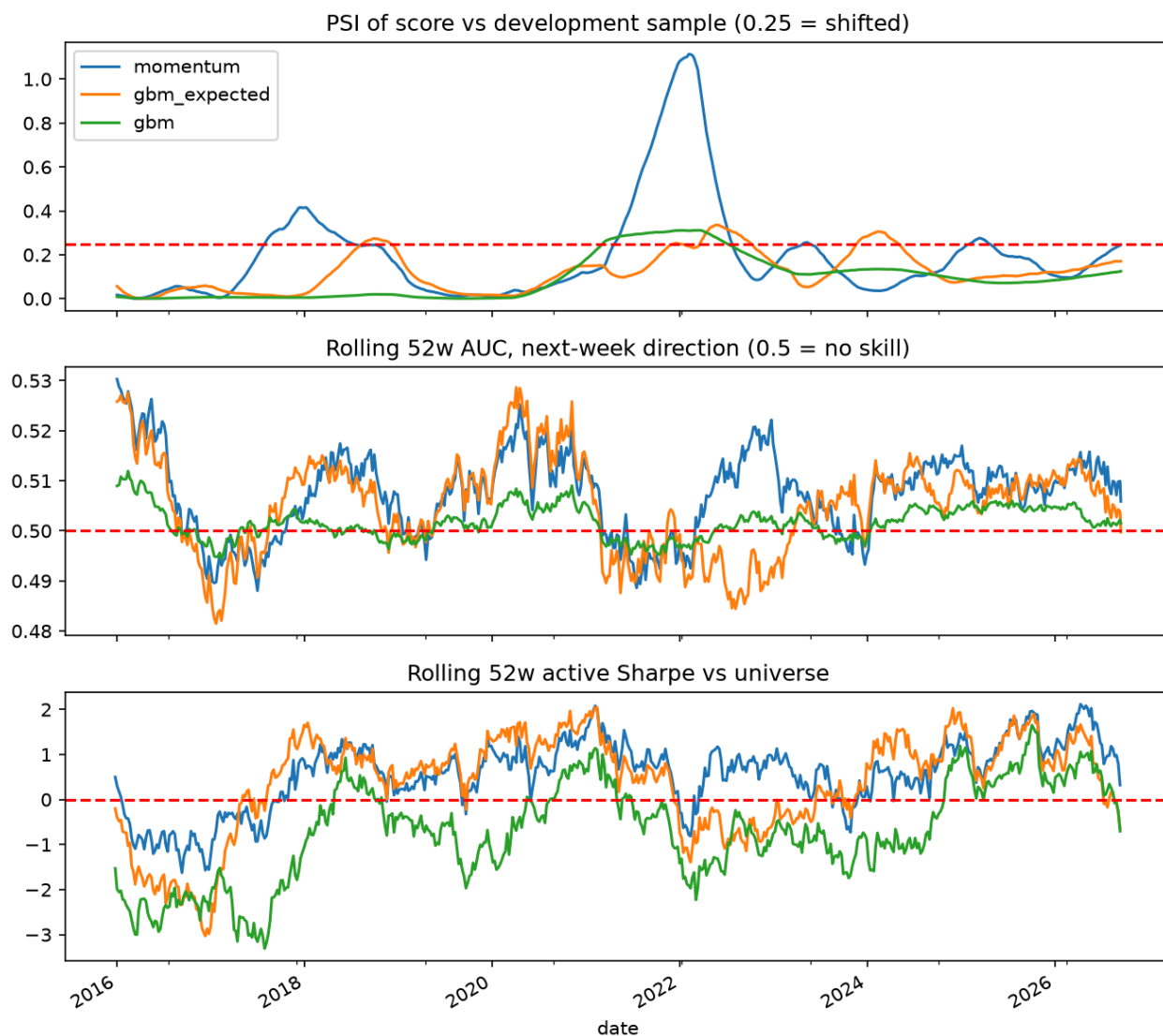
model	bootstrap SR [95% CI]	verdict	DSR	verdict	permutation p	verdict
momentum	0.65 [0.18, 1.10]	PASS	0.68	FAIL	0.006	PASS
gbm_expected	0.37 [−0.17, 0.89]	FAIL	0.31	FAIL	0.086	FAIL
gbm	−0.50 [−1.08, 0.05]	FAIL	0.00	FAIL	0.278	FAIL

With only three candidates, the expected maximum Sharpe of pure-noise strategies is already 0.51 annualised; momentum's 0.65 does not clear it with 95% confidence. This is the trading analogue of peeking in sequential testing: every additional candidate raises the bar.

6.3 Q2 — monitoring, latest reading (week of 2026-09-12)

metric	momentum	gbm_expected	gbm
psi_score	0.246 (watch)	0.171 (watch)	0.125 (watch)
psi_input (realised vol)	0.611 (shifted)	0.611	0.611
auc_1w	0.506	0.500	0.502
auc_13w	0.515	0.510	0.502
calib_slope (13w)	—	−0.03	0.01

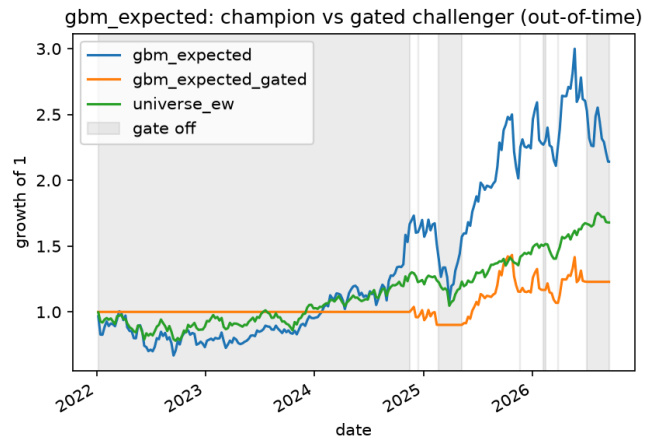
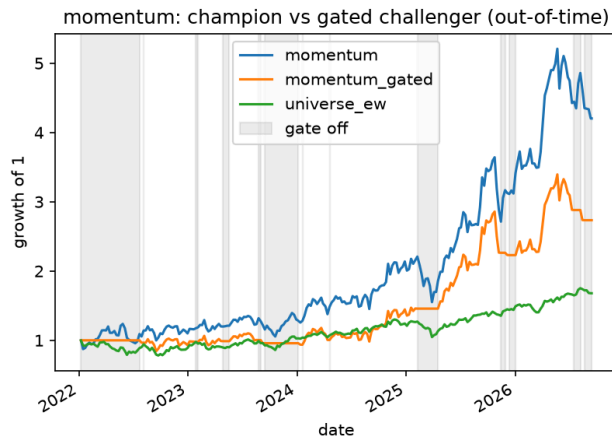
No model shows discriminatory power above 0.53 AUC in any calendar year 2015-2026 (SQL cross-check, `notebooks/02_monitoring.ipynb`). The predicted-return models are uncalibrated: a slope of −0.03 means a higher predicted 65-day return does not translate into a higher realised one. The volatility regime has shifted well beyond the development sample (PSI 0.61), which applies to all candidates equally.



6.4 Q3 — champion vs challenger, out-of-time 2022-01 → 2026-09

model	CAGR champion → challenger	max DD champion → challenger	Sharpe champion → challenger	weeks gated off
momentum	35.5% → 23.7%	-29.7% → -22.0%	1.02 → 0.88	48%
gbm_expected	17.5% → 4.4%	-37.5% → -25.4%	0.63 → 0.33	63%
gbm	5.9% → 0.0%	-34.7% → 0.0%	0.34 → n/a	92%
universe equal-weight	11.6%	-21.6%	0.67	—

The gate does what it is specified to do — it cuts maximum drawdown by 26% (momentum) and 32% (gbm_expected) — but it costs more return than it saves risk: Sharpe falls in both cases. The dominant trigger is active_drawdown, which fires late. As specified, the kill switch is a drawdown limiter, not a skill detector; a validator would not approve it as the sole control.



7. Findings

#	model	severity	finding	evidence
F1	gbm	High	Scoring by a single Monte-Carlo path adds noise of order $\sigma/\sqrt{T}$ that destroys the ranking: weekly turnover 90% vs 10%, CAGR 3.9% vs 18.8% for the identical model's expectation. The defect is in the implementation, not the model.	median weekly Spearman(single path, expectation) = 0.24
F2	gbm / gbm_expected	Medium	$E[\text{score}]$ depends on μ only and exp is monotone, so the "GBM simulation" ranks stocks by 2-year mean log return. The Monte-Carlo machinery adds no information.	Spearman(expected score, trailing mean log return) = 1.00
F3	clam_orig	Critical	Training sequences are cut from 94 tickers concatenated and sorted by date : a 252-row window covers ~2.7 calendar days of 94 different stocks, while inference feeds 252 days of one stock. The network was never trained on the input it receives in production.	code review of clam_model.py vs clam_inference.py
F4	clam_orig	Blocking	Weights dated 2025-08-22: every backtest week before that is in-sample. 56 out-of-time weeks remain, i.e. ~4 independent 65-day horizons — too few for any statistical statement.	panel date count after 2025-08-22
F5	momentum	High	Entering one session after the rebalance close removes roughly half the out-of-time CAGR (35.5% → 19.2%). The excess return is concentrated in the first session after each rebalance (top-50 first-session return 11–49% annualised in most years vs 4–8% for the universe). The backtest is only valid under close-of-rebalance execution.	ret_fwd_1w_lag1 sensitivity, notebooks/04_findings.ipynb
F6	all	Low	Yahoo returns negative adjusted closes for 3 tickers; without filtering,	notebooks/00_data_quality.ipynb

#	model	severity	finding	evidence
			momentum scores become infinite and monitoring crashes.	
F7	validator	Process	The original permutation test compared <i>net</i> returns and was biased toward passing any low-turnover signal. Corrected (§5.1).	§5.1

8. Limitations

1. **Survivorship bias** — unavoidable with free data; quantified in §4 and fig. 1. Absolute returns of every candidate are overstated; relative conclusions are more robust than absolute ones.
2. **Single universe snapshot** — market-cap ranks are as of 2026-09-15; the 2015 universe would have contained different names.
3. **Execution assumptions** — trades at the rebalance close at 10 bps one-way; no market impact, no borrow, no capacity constraints. F5 shows results are sensitive to this.
4. **Multiple-testing count** — the DSR uses $N = 3$ declared candidates. Any undocumented variants tried during development would raise the bar further.
5. **Deep-learning candidate** — `clam_orig` could not be validated on statistical grounds (F4) and `clam_2021` was not yet scored on the validation machine at the report date. The CLAM decision rests on code review (F3), which is sufficient for non-approval.
6. **Costs** are a flat assumption; small-cap names in the tradable set will trade wider than 10 bps.

9. Conditions of use and prohibited uses

Applies to `momentum`, the only conditionally approved model:

- Approved **as a benchmark** for evaluating future models, not as a standalone production signal, until the Deflated Sharpe test is passed on a longer sample or with fewer competing candidates.
- Use only within the tradable universe ($\text{close} \geq \$5$, $\text{ADV} \geq \$5\text{M}$). Scores outside it are undefined for approval purposes.
- Execution must occur at or within the rebalance session close. Any process that enters the following day is **out of scope** (F5).
- Do not rely on the kill switch as the sole risk control; it limits drawdown after the fact and reduced Sharpe out-of-time (§6.4).
- Not approved for horizons other than one week, for leverage, or for short positions.

Prohibited outright: `gbm` in its deployed single-path form (F1); `clam_orig` for any decision (F3, F4).

10. Ongoing monitoring plan

metric	frequency	amber	red	action on red
<code>psi_score</code> vs development sample	weekly	0.10	0.25	gate off; investigate feature drift
<code>auc_1w</code> (trailing 52w)	weekly	0.51	0.50	gate off; escalate after 4 consecutive weeks
<code>rolling_sharpe</code> (trailing 52w, active)	weekly	0.25	0.00	gate off
<code>active_drawdown</code>	weekly	−10%	−15%	gate off; review sizing
<code>psi_input</code> (volatility regime)	weekly	0.25	0.50	flag: all candidates are outside their development regime

metric	frequency	amber	red	action on red
full re-validation (Q1 tests)	annually, or after any code change	—	—	re-issue this report

All metrics are written to `monitoring` and `gate_decisions` in the DuckDB file by the same code that produced this report, so the monitoring pack is a query, not a re-implementation.

11. Recommendations to the model owner

1. **gbm**: replace the single Monte-Carlo path with the closed-form expectation (or many paths); then recognise via F2 that the model is a trailing-mean-return rank and decide whether that is the intended signal.
2. **clam**: rebuild training sequences per ticker so training and inference share an input structure; retrain with a fixed cut-off date and a held-out period of at least three years before resubmitting.
3. **momentum**: document the execution assumption; investigate the first-session concentration (F5) before any live use.
4. **Kill switch**: add a leading indicator (e.g. AUC deterioration with a shorter window) alongside the drawdown rule, and re-test champion vs challenger.

12. Round 2 — revalidation after redevelopment (2026-09-16)

Following §11, the model owner redeveloped both rejected models for a **weekly** horizon. The validator's role in round 2 was unchanged: same backtest engine, same three questions, same out-of-time window, and every variant the developer tried is counted as a trial in the Deflated Sharpe Ratio ($N_TRIALS_DSR = 10$: 3 round-1 candidates + 6 GBM variants + 1 CLAM variant retained after 3 attempts).

12.1 What the developer changed

model	change	developer-side selection (development sample < 2022 only)
<code>gbm_weekly</code>	horizon 65 → 5 days; the single Monte-Carlo path replaced by the closed-form expectation (F1); lookback re-chosen	6 variants (lookback 63 / 126 / 252 × expected / prob-up); best development Sharpe → <code>gbm_w_126_expected</code>
<code>clam_weekly_cs_demeaned</code>	sequences built per ticker (F3); training cut at 2021-12-31 (F4); horizon 5 days; scalar target = next-5-day log return minus that week's cross-sectional mean; top-500 training universe	validation rank IC 0.014
<code>clam_weekly_cs_rank_small</code>	as above with a within-week percentile-rank target and a smaller network	validation rank IC 0.016
<i>(discarded)</i>	raw-return target (IC −0.008); weekly-bar input (collapsed to a constant)	—

12.2 Results

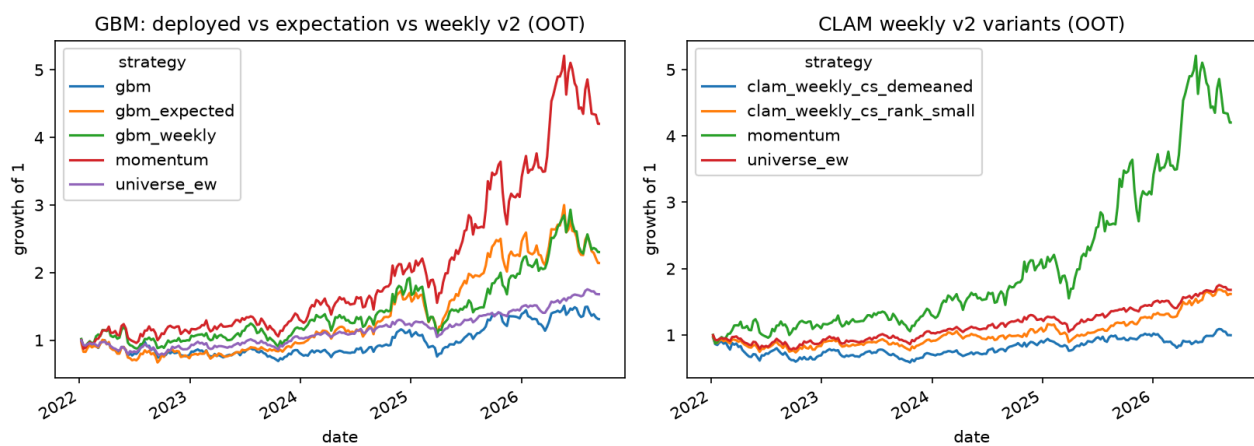
Q1, full sample 2015-01 → 2026-09, active return vs universe (DSR hurdle at $N = 10$ is $SR_0 = 0.86$):

model	active Sharpe [95% CI]	bootstrap	permutation p	DSR	passed
momentum (benchmark)	0.65 [0.18, 1.10]	PASS	0.004 PASS	0.24 FAIL	2 / 3
<code>gbm_weekly</code>	0.49 [CI > 0]	PASS	0.026 PASS	0.11 FAIL	2 / 3
<code>gbm_expected</code> (round 1)	0.37	FAIL	0.080 FAIL	0.05 FAIL	0 / 3
<code>gbm</code> (deployed)	−0.50	FAIL	0.226 FAIL	0.00 FAIL	0 / 3

model	active Sharpe [95% CI]	bootstrap	permutation p	DSR	passed
clam_weekly_cs_rank_small	0.04	FAIL	0.106 FAIL	0.00 FAIL	0 / 3
clam_weekly_cs_demeaned	-0.68	FAIL	0.704 FAIL	0.00 FAIL	0 / 3

Out-of-time 2022-01 → 2026-09:

strategy	CAGR	vol	Sharpe	max DD	one-day execution lag
momentum	35.5%	36.3%	1.02	-29.7%	19.2%
gbm_weekly	19.3%	36.2%	0.67	-43.4%	8.3%
gbm_expected	17.5%	35.6%	0.63	-37.5%	11.4%
clam_weekly_cs_rank_small	10.7%	22.5%	0.57	-24.1%	9.1%
clam_weekly_cs_demeaned	-0.1%	26.8%	0.13	-39.0%	—
universe equal-weight	11.6%	19.2%	0.67	-21.6%	—



Monitoring: no candidate exceeds AUC 0.53 in any calendar year (SQL cross-check); `gbm_weekly` is the only model with a meaningful calibration slope at the 13-week horizon (0.74). The gate switches `gbm_weekly` off 59% of weeks and turns its out-of-time CAGR from 19.3% to -2.6%, confirming the round-1 conclusion that the rule set as specified is not a usable control.

12.3 Findings added in round 2

#	model	severity	finding
F8	<code>gbm_weekly</code>	Medium	The fixed model is statistically distinguishable from noise (bootstrap and permutation pass) but its out-of-time Sharpe (0.67) equals the equal-weighted universe and trails the untuned benchmark (1.02) with a deeper drawdown. Reason for rejection moves from <i>implementation defect</i> to <i>no value over benchmark</i> .
F9	<code>clam_weekly_*</code>	High	After repairing F3/F4 and trying four target/input variants on a 478-ticker, 142k-window training set, the best validation rank IC is 0.016 and out-of-time performance is indistinguishable from the universe. The limitation is the approach — OHLCV sequences alone carry no exploitable weekly cross-sectional signal at this scale — not the implementation.
F10	process	Low	Multiple-testing accounting: raising the trial count from 3 to 10 lifts the DSR hurdle from $SR_0 = 0.51$ to 0.86 and fails every candidate including the benchmark. Development iterations must be logged so the validator can count them.

12.4 Decision

model	round 1	round 2
momentum	conditionally approved as benchmark	unchanged
gbm → gbm_weekly	not approved (F1)	not approved — passes noise tests, fails to beat benchmark (F8); may be resubmitted with a documented economic rationale and a smaller trial count
clam_orig → clam_weekly_*	not approved (F3, F4)	not approved — recommend discontinuing the sequence-model approach for weekly selection (F9)

Recommendation to the owner: stop iterating on CLAM; if GBM-weekly is pursued, its only demonstrated content is trailing mean return (F2), so it should be compared against — or merged into — the momentum benchmark rather than maintained as a separate model.

Appendix A — assumptions (config.py)

assumption	value
universe size / snapshot	3,000 / 2026-09-15
price history	2013-01-01 → 2026-09-14
backtest start / rebalance	2015-01-02 / weekly, last trading day (Friday)
holdings / weighting	top 50 / equal
cost	10 bps one-way on traded weight
liquidity filter	close ≥ \$5, 20-day ADV ≥ \$5M
momentum	252-day lookback, 21-day skip
GBM	504-day lookback, 65-day horizon
CLAM	252-day input, 65-day horizon; original weights through 2025-08-22; retrain cut 2021-12-31
bootstrap	stationary, mean block 13 weeks, 5,000 draws
DSR trials	3 (round 1) / 10 (round 2)
monitoring window / development sample	52 weeks / first 104 weeks
out-of-time window	from 2022-01-03

Appendix B — reproduction

README.md lists the eight commands that rebuild every table and number in this report from a fresh checkout. Workpapers: notebooks/00_data_quality ... 05_round2; developer side: Quant_Model_Research/clam_weekly.ipynb, Long_Term_Trading/gbm_weekly.py.